Arnav Maloo

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.94
Lord Buddha Public School, Kota	Class XII (CBSE)	2024	91.40%
Hind Zinc School, Chittorgarh	Class X (CBSE)	2022	95.40%

POSITIONS OF RESPONSIBILITY

Finance Head | Finance and Consulting Club, IIT Hyderabad

- **Directed** a 16-member team across sponsorship, live projects and content verticals, owning the club's annual roadmap and delivering initiatives in the tenure.
- Sourced live project pipelines by **pitching corporate partners**, structuring collaborative engagements that provided tangible industry exposure to students.

Finance Coordinator | Finance and Consulting Club, IIT Hyderabad

- Managed the **end-to-end product lifecycle** for large-scale competitive simulations (PortfolioX and Alpha Arena), defining platform requirements, capital constraints and user mechanics on Unstop and StockGro.
- **Instructed** intensive campus **workshops**, breaking down complex equity, fixed income, and valuation methodologies, including **DCF and LBO modeling.**

Internship Coordinator | Internship Cell, Office of Career Services (OCS)

- Coordinating between students, recruiters and the OCS team to facilitate internship-related processes.
- Assisting in internship drives, company outreach and communication with stakeholders.

PROJECTS

End-to-End Fraud Detection Pipeline | Python, pandas, NumPy, scikit-learn, CatBoost, LightGBM,SMOTE

- Built an end-to-end fraud detection pipeline on **20K transactions** with 1.7% fraud prevalence, using PR-AUC as the primary metric to avoid the accuracy trap of severe class imbalance.
- Benchmarked Logistic Regression, CatBoost and LightGBM under a stratified train-val-test split, showing the simpler linear model won on ranking (PR-AUC 0.38, ROC-AUC 0.94) and avoiding needless complexity.
- Designed a cost-sensitive decision rule ($200 false-negative vs $5 false-positive) evaluated under a fixed analyst alert budget, translating model scores into an operations-ready threshold.

DejaVu-NLP Pattern Discovery Engine | Python, FastAPI, React, Sentence Transformers, SQLite

- Built an **NLP pipeline** that finds **recurring concepts** across past **competitive-exam papers**, so prep targets what actually repeats, using **all-MiniLM-L6-v2 embeddings** behind a **FastAPI** backend and **React** frontend.
- Clustered questions with a **connected-components graph** approach that avoids fixing cluster counts upfront, tuning the **cosine-similarity** threshold against a synthetic ground-truth set to reach a **Pairwise F1 of 0.92**.
- Added a dashboard that turns clusters into frequency distributions across **2,000+ questions**, showing which concepts dominate the syllabus, so the dataset can be sampled by topic probability.

Cardmember Concierge - RAG support assistant | Python, FastAPI, Scikit-learn

- Built an AI support assistant for a credit-card company that answers customer questions by retrieving and citing from a knowledge base of **50+ help articles** and treats *knowing when not to answer* as the core problem.
- Designed a safety layer that automatically routes sensitive requests (fraud, disputes, legal) to human agents and made it a hard product constraint.
- Defined the success metrics (customer-safety, resolution rate, response speed) and built a live dashboard to track them, so the trade-off between automation and safety was measurable.

SKILLS

Finance: Financial Modelling(DCF,LBO), Valuation, Market Analysis, Comparable Company Analysis (Comps)

Business & Strategy: Growth Strategy, Go-to-Market (GTM), Market Entry, Competitive Analysis, Product Analytics, Profitability Analysis, Pricing Strategy, Business Case Development

Data & Experimentation: SQL, Data Analysis(Pandas,Numpy), A/B Testing, Hypothesis Validation

AI/ML: RAG, Scikit-learn, Pandas, NumPy, Sentence Transformers, NLP,Generative AI, LLMs

Technical Skills: Python, C++, Machine Learning(Regression, Classification, Clustering, Ensemble Learning, Feature Engineering, Hyperparameter Tuning, Cross Validation, Model Evaluation), Generative AI, JavaScript

RELEVANT COURSES

Programming Courses: Artificial Intelligence, Introduction to Programming, AI/ML Tool Chains

Mathematics: Applied Mathematics, Numerical Methods, Linear Algebra, Differential Equations

Others: Introduction to Entrepreneurship, Spoken Japanese

CERTIFICATIONS

- **Mastercard**(Advisors & Consulting Services), **BCG (**Strategy Consulting), **PwC (**Management Consulting) | Job Simulations
- **Business & Sponsorship** Case Study (FC Barcelona) **|** prooV
- **Generative AI, Data Science & Machine Learning |** Krish Naik

SCHOLASTIC ACHIEVEMENTS

- Delivered a comprehensive **2-day Finance workshop to 150+ students** covering Markets, Investments, Valuation and guiding them through a **hands-on** DCF Valuation.
- Solved **400+ questions** across platforms like Leetcode, GeeksForGeeks & Coding Ninjas